UMEÅ UNIVERSITY

Monthly Electricity Consumption Forecasting Using an Explainable AI Framework

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Abstract

Checklist for final touches:

- Fix placing of figures and tables
- Clean up references
- (Center page content?)

Acknowledgements

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1 Introduction

In recent years, the Swedish electricity market has been affected by large variations in price and periodically high electricity costs[1][2]. This has increased the need for customers to understand and influence their electricity consumption and future electricity costs.

With the emergence of environmental problems due to climate change, sustainable energy management is becoming increasingly important [1, 2]. The ability for customers to understand and influence their household energy consumption is significant for sustainable energy use. Household energy consumption makes up a considerable part of the total energy consumption, approximately 30% in some European and American countries. Managing household energy more efficiently has large potential in saving energy, since it is estimated that 27% of energy used by households can be saved through more efficient use [3].

[Something about how the energy demand in the world is increasing]

[Something about how forecasting electricity consumption can help users make smart decisions about their consumption, both saving money and benefitting the environment.]

[Something about monthly forecasting and why that is useful]

For users, the ability to have confidence in the forecast is important. Users may, for example, question how the forecasting mechanism makes decisions, or question what factor is the most determining in the forecast. A system that can explain such things to the user is able to increase its reliability with users. Handling users' data in a safe and transparent manner is also important when implementing Artificial Intelligence (AI) based systems. Citizens of the European Union have a right to transparency and information about the decision-making of AI models that have a direct link to them, as outlined in the General Data Protection Regulation [4]. [Something about how XAI helps with these things]

Previous research has identified various models for predicting electricity consumption, both statistical and AI based [5, 6]. In recent years, increasing interest has been shown for AI solutions to energy consumption forecasting, due to their strength in handling nonlinear problems [4, 5]. Forecasts are often based on several factors such as weather conditions and building conditions, causing nonlinear patterns that statistical methods struggle with [5]. However, one disadvantage with AI methods is the black-box nature of the models which causes a lack of transparency and interpretability in the decision-making process. To solve the problem of elucidating the internal process and decision-making of AI algorithms, the field of explainable AI (XAI) has emerged within recent years. XAI methods are able to open up the black box, and provide explanations for why a model predicts a certain way [7].

[1] Allhorn, J., Tidningen Näringslivet: Chockvändningen: Nu har norra Sverige högst elpriser - "Har förändrats ganska snabbt" (January 2026). https://www.tn.se/inrikes/46369/chockvandningen-nu-har-norra-sverige-hogst-elpriser-har-forandrats-ganska-snabbt/

[2] Rydegran, E., Energiföretagen: Dramatik och rekord sammanfattar Elåret 2022 (December 2022). https://www.energiforetagen.se/pressrum/pressmeddelanden/2022/Dramatik-och-rekord-sammanfattar-Elaret-2022/

[To address this need, Umeå Energi has worked on developing and improving their interfaces for customers, for example by developing an application for private customers. According to customer analytics from Umeå Energi, a highly requested feature is to receive an estimation of both the electricity consumption and the electricity cost before the invoice arrives. Currently, Umeå Energi has developed a linear regression model to forecast the consumption for the current month. The model is based on historical consumption and temperature, aggregated by month. For certain customer groups, Umeå Energi has access to high definition measurement data where customer electricity consumption is registered on a 15-minute level, which creates opportunities for more advanced modeling. There may also be other factors, apart from historical consumption, that affect the electricity consumption, for example weather data, calendar information (holidays etc.), and energy prices.]

[This Master's thesis will investigate and quantify the contributions of different variables to a monthly electricity consumption forecast. Previous research has identified several models that are able to accurately predict energy consumption for different time intervals, utilizing various combinations of data sources [6, 5]. In recent years, increasing interest has been shown for Machine Learning (ML) and AI solutions to energy consumption forecasting, due to their strength in handling nonlinear problems [4, 5]. However, the black-box property of many popular AI models has increased the demand of asserting confidence and transparency through means of Explainable Artificial Intelligence (XAI) [4]. This study will address this demand by using XAI methods to evaluate how different variables such as weather data, holidays, and energy prices affect the accuracy of monthly household electricity consumption forecasts. Previous research has used XAI approaches for short-term forecasting (hourly-daily), but this study will explore a novel monthly scope [2, 8, 4, 9]. The access to customer-level data in this study also provides an aspect that is underrepresented in previous works [5, 10, 9]. **RQ1: Which forecasting variables contribute significantly to the accuracy of the monthly forecast, and which variables provide little to no improvement?**

To perform the forecasting, state-of-the-art ensemble learning models will be explored. Previous studies have used ensemble learning for short-term (hourly-daily) forecasting [11, 12, 9], but this study will address the novel task of monthly forecasts using ensemble learning. **RQ2: Which ensemble learning models are the most efficient for forecasting monthly electricity consumption?**]

For Umeå Energi, this research will add value by supporting customers in making informed decision about their energy consumption. According to internal customer analytics, a highly requested feature is to receive an estimation of both the electricity consumption and the electricity cost before the invoice arrives. Umeå Energi is currently in the process of launching an application for private customers. The ultimate goal is to include to monthly consumption forecasts in the application.

[Definition of short/mid/long-term forecasts]

[The contributions of this work can be summarized as follows:]

2 Theory

This chapter presents relevant theoretical background for the concepts and methodologies used in this research.

2.1 Time Series Forecasting

A time series is a set of chronologically arranged observations, generally under the assumption that time is a discrete variable [13]. In a time series with T real value samples $x_1, \ldots, x_T$, the value at time i is represented by x_i $(1 \leq i \leq T)$. The problem of time series forecasting can be defined as predicting the values of $x_{w+1}, \ldots, x_{w+h}$, where w is the historical window, i.e., the number of values considered in order to produce the prediction, and h is the prediction horizon, i.e, the number of future values to be predicted. Prediction happens given the previous samples $x_1, \ldots, x_w$ $(w + h \leq T)$, with the goal of minimizing the error between the predicted value $\hat{x}_{w+i}$ and the actual value x_{w+i} $(1 \leq i \leq h)$ [11].

A time series may be univariate or multivariate, where univariate refers to a series with a single observation recorded over time, and multivariate refers to a series with a group of variables with interactions. Analysis of time series often begins with obtaining the underlying patterns of the observed data. The next step is fitting a model to represent the data for future predictions, which can be a complex task. The analysis of the time series is usually performed by decomposing it into three components [6, 14]:

1. *Trend* - The general movement of the variable during the observation period. Trend does not take irregularities and seasonality into account.
2. *Seasonality* - The periodic fluctuation of the variable. It includes stable effects, along with time, magnitude, and direction.
3. *Residual* - The remaining, largely unexplainable part of the time series. In some instances, this can be high enough to mask the trend and seasonality.

Both linear and nonlinear techniques may be used for solving time series problems. In linear methods, the idea is that strong correlations in the data allow for linear combinations to determine the next observation. However, the random component of the time series may prevent precise predictions. Nonlinear methods are applied in the machine learning domain to forecast future values based on a model describing the data. Machine learning solutions to forecasting problems have gained popularity, due to the fact that they are suitable both for linear and nonlinear problems [11]. Time series analysis has been extensively applied to the problem of energy consumption forecasting, especially since real-time monitoring of buildings has become more common, and the availability of recorded data has increased [6].

The scope of energy consumption forecasting problems can generally be divided into three categories; short-term forecasting (1 hour to 1 week), medium-term forecasting (1 month to 1 year), and long-term forecasting (1 year and above) [5]. [maybe move this to introduction for context]

2.2 Ensemble Learning

Ensemble learning is a family of methods that combine several machine learning algorithms to make decisions. It can be seen as the ML interpretation of "wisdom of the crowd", that is, the idea that aggregating the opinions of several individuals is better than selecting the opinion of one individual. Multiple weak learners make predictive results that are fused together via voting mechanisms, in order to achieve better performance than from any single algorithm. Any type of machine learning algorithm, e.g. decision tree, neural network, or regression model, can be used as an ensemble learner [15, 16].

[maybe a diagram giving an overview of the ensemble "steps"]

According to Dietterich [17], there are three reasons why ensemble learning methods achieve good performance in machine learning. The first reason is statistical. Models can be seen as searching for the best hypothesis within a hypothesis space H. When data is limited, which it often is, the models can find several hypotheses in H that give the same accuracy on the training data. This makes it hard to choose between hypotheses, since it is unknown which one will generalize better for unseen data. Ensemble learning can help with avoiding this problem of overfitting. Secondly, ensemble learning provides computational benefits. Many models function via performing some type of local search, which may be prone to getting stuck in a local optima. A better approximation of the unknown true function may be obtained from an ensemble constructed by the local search started from many different points. The third reason is representational. In most situations, the unknown true function may not be included in H. Combining several hypotheses from H, the space of representable functions can be expanded to possibly include the true function.

Three common strategies for constructing ensembles are bagging, boosting, and stacking [9]. Bagging, or bootstrap aggregating, proposed by Breiman [18] is a method for obtaining an aggregated prediction by building multiple versions of a predictor. Bootstrapping is used to generate multiple samples of data from the original dataset, which are then used to independently train the different predictors. Majority voting is used to aggregate the final prediction [15]. The main objective of the bagging technique is to reduce the variance of a prediction model. A common example of the bagging technique is the random forest (RF) algorithm [9, 19].

In boosting, several models with individually weak predictions are aggregated to obtain a model with high accuracy. Models are trained sequentially to correct the errors of previous models. The main objective with this technique is reducing model bias [9, 19, 20]. Two examples of boosting algorithms are Gradient Boosting Machines (GBM), which uses gradient descent to minimize the loss function, and XGBoost, which is an extension of GBM focused on resource optimization and prevention of overfitting [19].

Stacking utilizes different learning algorithms called base learners to make initial predictions. These outputs are then combined via a meta-learner to make the ultimate predictions.

The goal is to capture a wider range of patterns using the strength of different models, while minimizing the generalization errors [9, 19, 11]. Accuracy improvements mainly happen when there is diversity among the individual models, since differences in generalization principles generally lead to different results [19].

2.3 Explainable Artificial Intelligence

In recent years, artificial intelligence has been widely adopted in society, and humans increasingly rely on AI as part of decision-making. Research in the field of AI has for many years focused on improving the predictive accuracy. However, the black-box nature of many AI systems creates a lack of transparency, and explaining to users how the AI systems work is highly problematic. Pressure from society, ethics, and legislation has created a demand of a new type of AI that is interpretable to users and can explain its inner functions [7, 21].

According to Saarela and Jauhiainen [22], there are two main objectives with classifiers; they should be able to make accurate predictions for new given input features, and they should be interpretable (i.e., provide explanations for how a given input relates to the output). Usually, models have a trade-off between accuracy and interpretability. Linear regression models are generally transparent and easy to understand, but their performance is often worse than more complex nonlinear models. On the other hand, the nonlinear models perform well but lack in interpretability.

Explainable AI (XAI) methods that elucidate the decision-making process have been developed to enhance the interpretability of AI models. These methods can be applied at different levels of the AI modelling. Firstly, explainability can be applied pre-modelling through means of data pre-processing and data analysis. This approach gives insights into the datasets used for training the ML models, and it can expose imbalances in the data. Secondly, the model itself can be inherently interpretable. This refers to models that have a simple structure, such as linear regression and k-Nearest Neighbors. Models like these are understandable by design, and users are able to interpret the model based on the model summary or parameters. Thirdly, explainability can be applied post-modelling to deal with the black-box problem of more complex ML models. Post-modelling methods include textual and visual justifications, simplification of the model, and feature relevance. These methods can explain how an algorithm performs during training, and how predictions are generated given a certain input. XAI approaches may be model-specific, meaning they are applied to a specific type of model, or model-agnostic, meaning they are not dependent on model architecture and can be applied for any model [7, 21].

Feature importance is the most common explanation tool for classification models [22]. It describes how important a variable was for the model output, that is, the individual contribution of a feature, regardless of the shape or direction of the feature's effect, dependent on a particular classifier. Feature importance reveals which features are crucial for the model when making predictions. The feature importance descriptions may be local or global, where local importance describes the feature's contribution to a specific prediction, and global importance describes the feature's contribution for the entire model [22, 21, 7].

Ensemble models, used in this research, are based on the aggregation of several models, which makes it more complex to interpret the results. Since the models are not interpretable by design, there is a need for the implementation of post-modeling XAI methods to understand the models' predictions. Among the post-modeling XAI methods, simplification and feature relevance are the most widely used for this task [7]. [Give examples of how these have been applied (?)]

[Maybe have a separate section for local/global explanations. And model agnostic/model specific explanations.]

2.3.1 SHAP

SHapley Additive exPlanations (SHAP) is a popular method for model explainability. It is model-agnostic, meaning that it can be used for any ML model. The SHAP method has its foundation in game theory, where the contribution of each player to the total payoff is calculated. In the case of machine learning, the players are represented by features (variables) and the payoff is the model output. To calculate the contribution score for each feature, all different combinations of features (i.e., coalitions in game theory) are evaluated. The importance of feature i, I_i (for $i = 1, \ldots, n$), can be calculated according to Equation 2.1, where $\phi_i^{(j)}$ is the marginal contribution of feature i compared to j. SHAP can be used to explain model outputs both locally and globally, i.e., both for specific instances and for all instances [23, 10].

$$I_i = \sum \left| \phi_i^{(j)} \right| \tag{2.1}$$

When using SHAP for model explanations, it is important that users are aware of some important considerations of the method. Firstly, the SHAP method is model-dependent, meaning it can produce different explanations for the same task when different models are applied. Secondly, the explainability scores assigned to features do not represent the weights of the features. Rather, the explainability lies in the ranking of the features, meaning that feature importance is deduced from the ordering. Thirdly, it is assumed that the features are independent of each other. Therefore, some features that are correlated with other features may receive low scores even though they contribute significantly to the output. This is because the correlated features have already accounted for the impact of the low-scoring feature. Lastly, if SHAP is applied to a biased classifier (e.g. due to data bias or algorithmic bias), the method may produce unrealistic explanations that do not describe the bias. Because of these points, it is crucial to present SHAP results in an informative way by including corresponding output plots and assumptions behind the method [23].

[maybe include the four properties of shap/Shapley values (specifically in an ML context): efficiency, symmetry, dummy player, additivity (but need source, https://machinelearningmastery.com/a-gentle-introduction-to-shap-for-tree-based-models/)]

3 Previous Work

Research in the field of electricity consumption forecasting has established many methods for producing forecasts. These methods can roughly be divided into two categories; conventional methods and AI-based methods. Conventional methods such as time series analysis and regression methods have previously been widely applied to solve the problem of electricity consumption forecasting. Nowadays, AI-based methods are more popular due to their strength in solving nonlinear problems, something that the conventional methods struggle with [5].

[Two comprehensive reviews on electricity consumption forecasting have been identified:

- An extensive review and comparison of both statistical and ML/DL techniques for forecasting is found in [6]. They also review combinations of different techniques, i.e. hybrid models. Claims that ANN has more advantages than statistical models, and has better performance for nonlinear problems. Highlights that hybrid models can be beneficial to capture complexities in building energy and operational data.
- Another review of statistical, AI, and hybrid methods for forecasting is found in [5]. They also highlight the strength in AI models for dealing with nonlinear patterns. Claims that hybrids between AI and Swarm Intelligence (SI) methods show potential for increased accuracy. Provides a clear overview of different studies regarding prediction time intervals, included features, building types etc.]

3.1 Machine Learning Approaches?

[Forecasting daily cooling energy using ANN (for three university buildings): [24]]

3.2 Monthly Forecasting

[Could possibly use [25] as an "early" example of monthly forecasting. They use SVM, but do not handle seasonality in any certain way.]

Forecasting electricity consumption in the monthly scope requires handling complex nonlinear patterns and seasonal tendencies [26]. Monthly consumption is affected by factors such as economy, temperature, and season, which makes the consumption exhibit a mixture of trend, periodicity, and randomness [27]. Additionally, when predicting the consumption of small-scale users, like in this thesis, the power demand is more sensitive to random factors. Human behavior may change periodically, both due to social factors such as holidays and summer vacations, and due to natural factors like the seasons [28].

To address the seasonality of monthly forecasting, models incorporating seasonal indexing (SI) have been proposed. Cao and Wu [26] aimed to improve monthly electricity consumption forecasts using support vector regression (SVR) in combination with SI and a novel evolutionary algorithm called fruit fly optimization (FOA). They calculated the seasonal index for each month, which was then used to adjust the predicted values according to the seasonal tendencies. FOA was applied to find the best values for parameters of the SVR model. Their results showed that both SI and FOA was able to effectively improve the forecasting accuracy, suggesting that the hybrid approach is a viable option for monthly forecasting.

Tang et al. [27] proposed a model specifically designed to handle sparse data with periodic properties. Their model also incorporated SI to address the seasonal component. A GM(1,1) gray model, a mathematical optimization model for incomplete data, was used to predict the seasonal index. The SI was combined with a trend forecast value to obtain the final prediction, and results showed that the model significantly improved accuracy, even for small electricity consumption datasets.

Sun et al. [28] approached the monthly forecasting problem by seasonal and trend decomposition of the electricity load time series, in combination with Auto-regressive Integrated Moving Average (ARIMA). This technique allowed for control of both the seasonal component's change rate and the trend component's smoothness, as well as increased robustness to outliers. Additionally, they incorporated economic factors into the forecast to improve accuracy. They found that the proposed model was able to increase forecasting accuracy, and determined that both the decomposition approach and inclusion of economic variables were effective to the forecast.

3.3 Ensemble Learning Approaches

Ensemble learning has previously been applied for the task of electricity consumption forecasting, mostly in the short-term scope. Divina et al. [11] proposed a stacking ensemble learning approach for short-term forecasting of Spain's general electricity consumption. They used a two-layer approach based on a bottom and a top layer. The bottom layer consisted of three base learning methods; regression trees based on evolutionary algorithms, Artificial Neural Network (ANN), and Random Forest (RF). The top layer used Generalized Boosted Regression Models (GBM) to combine the predictions of the base learners, and produce the final predictions. They found that the combination of weaker learning methods was able to produce superior results compared to several baselines, showing that ensemble learning is suitable for short-term electricity consumption forecasting. Neural networks were able to predict with high accuracy in the very near future, but accuracy declined when predicting further in the future. The ensemble method also showed significantly better performance compared to baselines when the amount of historical data was small, indicating that ensemble methods may be more flexible for different quantities of data.

Pinto et al. [12] also utilized ensemble learning for forecasting electricity consumption in the short-term. Using consumption data from an office building, along with other variables such as temperature and humidity, they predicted hour-ahead consumption with three different ensemble methods. These methods were random forests, gradient boosted regression trees (GBR), and AdaBoost. They found that ensemble learning methods, especially

AdaBoost, outperformed benchmark models such as support vector regression and fuzzy rule-based methods. They highlight including temperature and humidity information as advantageous to the forecasts, and they suggest that future work should consider additional external variables such as solar irradiation and thermal sensation.

Khan et al. [29] presented a stacking deep learning ensemble for short-term forecasts. The lowest level of the ensemble was compromised of multiple individual Long Short Term Memory (LSTM) models responsible for producing outputs. The top level of the ensemble was a Gated Recurrent Unit (GRU) responsible for increasing accuracy by combining outputs from the previous layer. They also utilized clustering to identify customer groups with similar profiles and load patterns. Based on hourly residential consumption data, they predicted hourly, daily, and weekly consumption, and found that the proposed model outperformed benchmark deep learning models. The clustering of customers helped with reducing model errors. Additionally, it was found that their model performed best for hour-ahead forecasts, while errors increased when approaching the weekly scales.

Hadjout et al. [30] used an ensemble deep learning approach to forecast monthly electricity consumption for high voltage customers in the Algerian economic sector. They utilized three different deep learning models - LSTM, GRU, and Temporal Convolutional Networks (TCN) - and grid search to find optimal weights for each model in the final ensemble prediction. Compared to the individual models separately, the ensemble approach performed significantly better according to MAE and MAPE metrics.

Long-term forecasting using ensemble learning has been researched by Chen et al. [31]. They predicted the annual electricity consumption for American households using extreme gradient boosting forests and deep networks as base learners, and ridge regression as a combiner method. The data included variables pertaining to household demographics, household type, appliances, and geographical data, providing a wide range of information. The ensemble framework was able to reduce errors by 30% compared to classical regression models such as linear regression and support vector machines. Investigation into the importance of input features using principal component analysis showed that electricity price, number of rooms, natural gas usage, and temperature during summer were among the variables that contributed the most information. They also note that some features, like number of rooms, number of bedrooms, and total square footage, have a strong correlation and have potential to be represented more efficiently. They argue that feature selection is necessary, since discarding of unimportant features can enhance model performance.

3.4 Explainable AI Approaches

Explainable AI methods have previously been applied for the task of short-term electricity consumption forecasting. Sim et al. [2] proposed a novel XAI approach for variable selection in energy consumption forecasting. They analyzed the influence of 15 different input variables pertaining to time information, climate data, and past energy consumption data. LSTM was selected as prediction model and SHAP was used to interpret the variables. The findings showed that time data such as year and hour had a strong influence on the prediction results, along with temperature, surface temperature, and energy-consumption-difference. Variables with a weak or ambiguous influence included humidity, wind speed, and sunshine amount. They concluded that forecasting performance can be significantly improved by selecting high-impact variables based on XAI analysis.

Maarif et al. [8] investigated the effect of 9 different features on energy consumption forecasting for a South Korean steel company. The features provided information about historical consumption, lagging and leading current power, carbon dioxide emissions, and load type. They utilized various LSTM models to make hour-ahead forecasts, and they used SHAP for the XAI analysis. It was found that leading current reactive power and seconds from midnight strongly influenced the model output, while CO_2 and load type had no effect on the forecast. They note that it is important to consider interpretability and performance trade-offs when developing XAI forecasting models, and that further research into this topic is needed for understanding the applicability of these models in the energy sector.

Januja et al. [4] maybe, but not reputable journal.

Another hourly electricity consumption forecasting study was performed by Lazzari et al. [10]. They provided a novel perspective by incorporating user behavior to enhance the accuracy of residential energy consumption forecasting. User behavior clusters were identified, and the day-ahead behavior was predicted using an XGBoost model. The predicted behavior, along with historical consumption, was then provided to an ANN model responsible for predicting the hourly day-ahead consumption. Compared to an ANN without the user behavior input, the proposed model was able to reduce MAPE and NRMSE by 7% and 9% respectively. A feature importance analysis using SHAP showed that calendar variables, namely week and season, had the greatest importance for the model. Temperature, notably minimum and maximum daily temperature, also had a large influence on model output.

Moon et al. [9] utilized ensemble learning and XAI methods for short-term residential building electricity consumption forecasting. They performed an extensive comparison between five ensemble learning models and ten deep learning models. It was found that the decision tree-based ensemble methods were superior in handling diverse and noisy data, rendering them useful for electricity consumption forecasting. They highlight robustness, interpretability, and efficiency as advantageous characteristics in ensemble models. They argue that ensemble models can handle data that is not extensively preprocessed and that they require less data than deep learning models. A feature importance and SHAP analysis was performed on several ensemble models, revealing that temperature-humidity index and wind chill temperature had significant importance for the forecasting. The SHAP analysis combined with the decision tree-based models provided interpretive insights about complex interactions in the data.

XAI approaches have also been applied to long-term forecasting [32]. Chakraborty et al. [33] predicted long-term cooling energy consumption for residential and commercial buildings under different climate change scenarios. They used SHAP to explain the results and concluded that the XAI method increased the interpretability of the XGBoost prediction model. Another long-term study was conducted by Wenninger et al. [34], who predicted the annual energy consumption for German households, and proposed a novel QLattice model for enhanced explainability. They found that the QLattice model was appropriate for the task of energy consumption forecasting, even though XGBoost achieved slightly better predictive performance. However, a significant strength with the QLattice algorithm is its transparent design, which was found to greatly increase the model explainability compared to established models such as ANN, XGBoost, and Support Vector Regression (SVR).

[With this, it is clear that there is a gap in research on XAI for medium-term forecasting.

And in general, the consensus is that more research in the XAI field in the energy sector is needed (?)]

4 Methodology

TODO list:

- Should change entire method to past tense
- It is possible to make diagrams for the decomposition of the time series [33].
- Maybe relevant to measure time for training models?

This study aims to develop monthly electricity consumption forecasts for residential households, with a focus on forecast explainability. A variety of input variables, including historical electricity consumption, household information, weather variables, and calendar information, are used to provide diverse information to the forecasting models. Five decision tree-based ensemble learning models are utilized for the forecasting, and their predictive performance as well as their explainability is assessed and compared. The explainable AI method *SHAP* is used to investigate the effect and importance of the model features. Figure 1 shows the general architecture of building the explainable electricity consumption forecasts.

4.1 Data Preparation and Analysis

This study utilizes electricity consumption data from Umeå Energi, along with external variables in the form of weather data and holiday information. This section describes the data, presents analytical insights as part of enhancing the explainability of the forecasts, and details how the data is preprocessed before performing forecasts.

4.1.1 Electricity Consumption Data

The Umeå Energi dataset contains monthly electricity consumption from 14 286 Swedish households, consisting of 4 years of measurements retrieved from January 1st 2022 to November 30th 2025. This data is provided by Swedish electricity company Umeå Energi. For the purpose of this research, households are defined as unique customer and facility combinations (i.e., one customer may have several facilities, which are regarded as separate households). Households that have any undefined or less than 10 kWh consumption measurements during the time period are removed from the dataset, leaving 13 692 households with complete measurements. All customers are private customers located in Umeå in the northern part of Sweden, meaning the data represents household electricity consumption in a cold climate. The dataset is summarized in Table 1, and Figure 2 shows the distribution of the electricity consumption for all households during the entire time period.

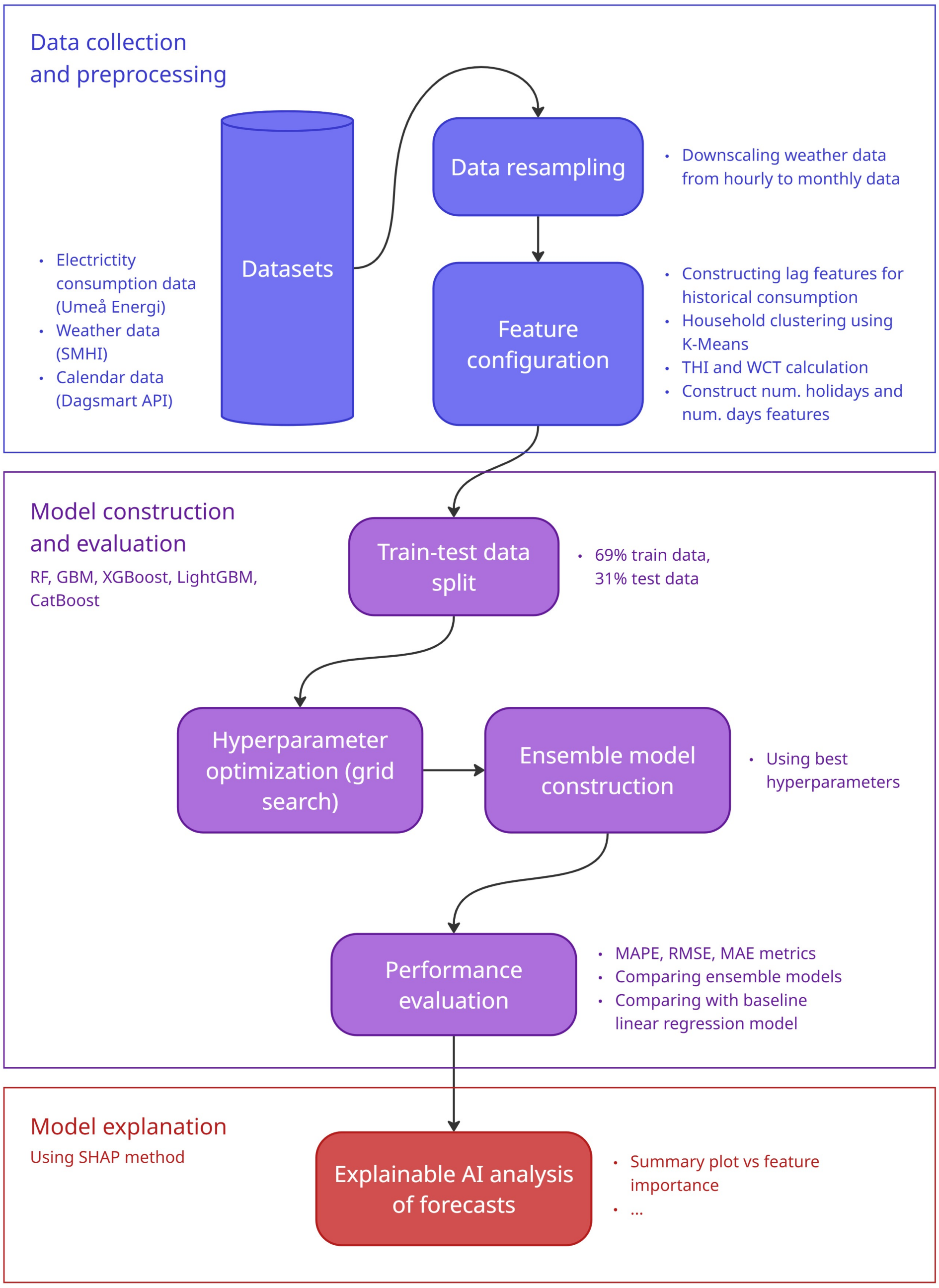


Figure 1: Architecture of modelling explainable electricity consumption forecasts [fix this figure].

Table 1 Summary of the Umeå Energi electricity consumption data.

Statistic	**Umeå Energi Data (kWh)**
Minimum	10
First quartile	124
Median	220
Mean	465
Third quartile	489
Maximum	13 490
Time range	2022-01-01 to 2025-11-30
Location	Umeå, Sweden
Number of households	13 692

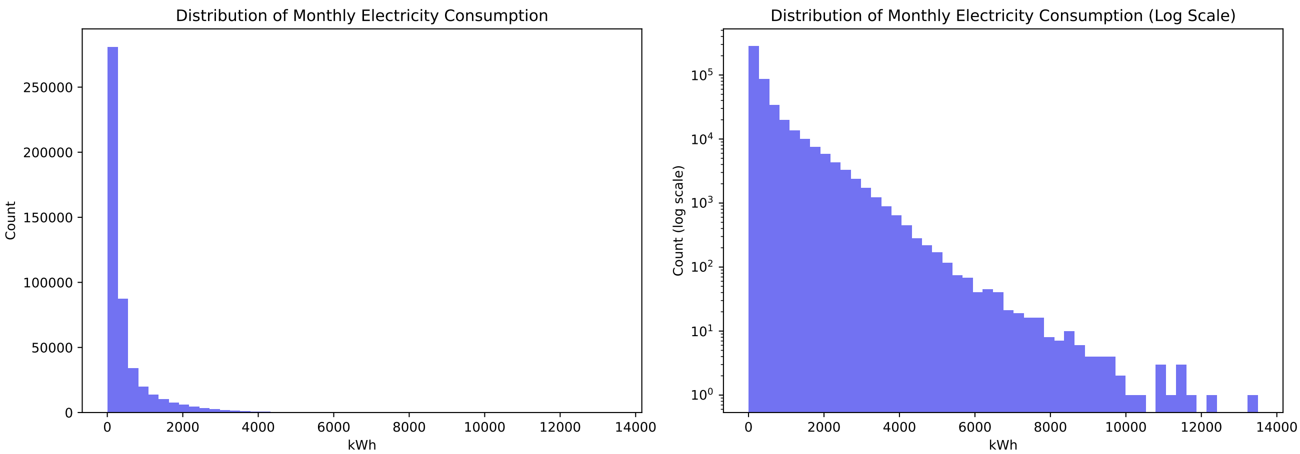


Figure 2: Histogram over the monthly electricity consumption distribution from January 2022 to November 2025.

Figure 2 and Table 1 show that a vast majority of households consume electricity in a similar low range per month, but that there are outliers where the monthly consumption exceeds several thousand kWh. Figure 3 shows the average monthly consumption for all households. It is clear that there is a significant seasonal pattern in the consumption, with more electricity being consumed in the winter and less in the summer. This makes sense for the cold climate, since the winter requires more heating for households with electrical heating, while summers are cool enough to not require much electrical cooling. A warmer climate may have the opposite consumption pattern, since warm summers will have a higher need for cooling measures [source]. Figure 4 illustrates the average consumption by month, showing that consumption varies more during the winter months, while the average consumption is more stable during the summer.

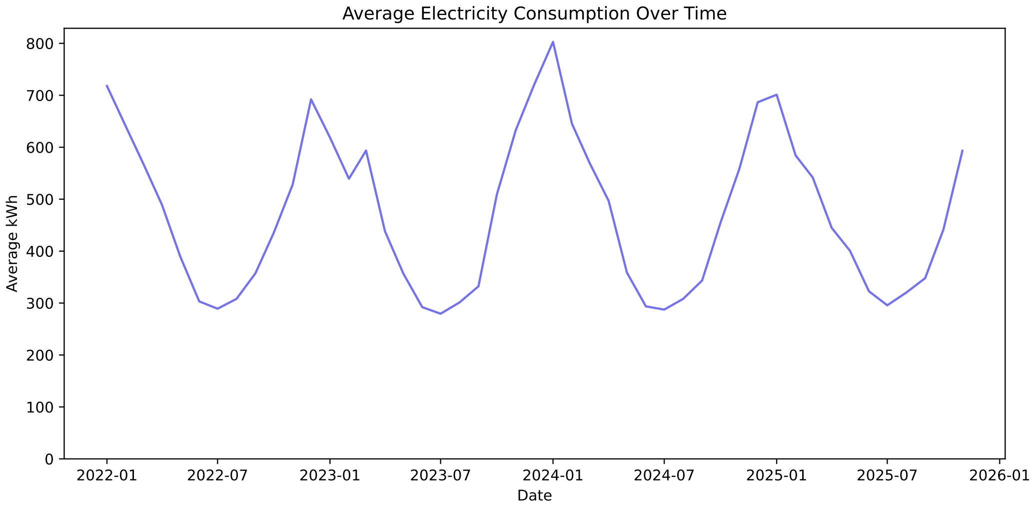


Figure 3: Average household electricity consumption from January 2022 to November 2025.

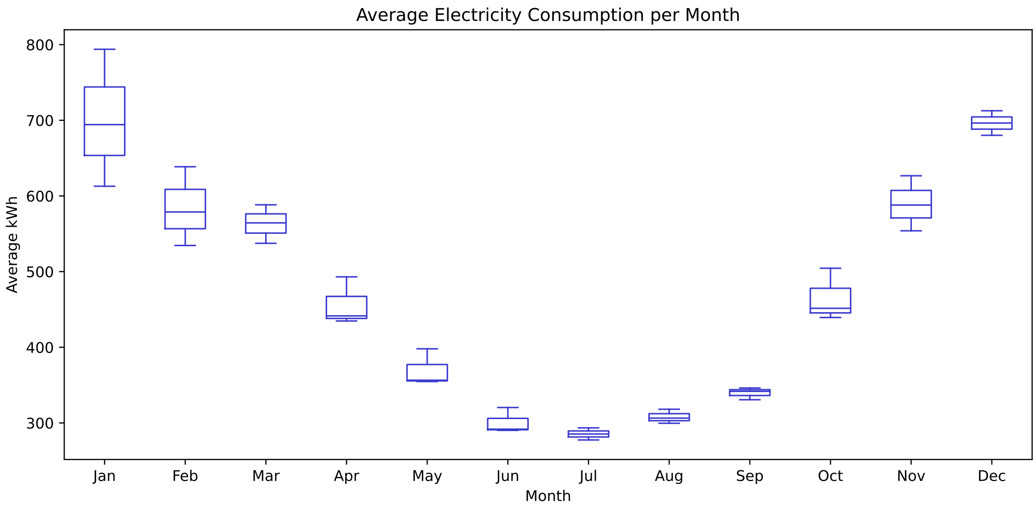


Figure 4: Average household electricity consumption by month from January 2022 to November 2025.

Three input features are constructed from the historical electricity consumption; lag_1 which represents the consumption of the previous month for the target household, lag_12 which represents the consumption for the same month the previous year for the target household, and $rolling\_mean\_3$ which represents the average consumption over the past three months for the target household. These variables express the recent consumption patterns, but also the monthly dynamic. Since the year 2022 does not have any past year historical consumption to refer to, it is removed from the training data.

Household Information

In addition to the historical electricity consumption, the Umeå Energi dataset contains information about the households. Households are labeled by type of residence, type of heating system, and type of contract agreement with Umeå Energi. The type of residence is either house, apartment, or vacation home. Heating can be electrical, non-electrical, or district heating. Contract agreements may be variable, fixed, or default assigned contracts. The share of households in each category is found in Table 2. Any given household may change between categories over time (e.g., change its type of heating system), and therefore, the household information is updated per month. Subsequently, Table 2 is not presented in a number of households per category, but rather, the share of observations that occupy a given category.

These features are not directly maintained by Umeå Energi, and instead, the labeling is done by Umeå Energi based on certain properties of the customer, which may cause the labeling to be inaccurate in some cases. However, it provides a good estimation of the household properties. In order to use the properties as input to the forecasting models, the string values are encoded as integers [except I should not do it for CatBoost].

Table 2 Household types in the Umeå Energi dataset.

Type	**Percentage**
Apartment	58%
House	38%
Vacation home	4%
Non-electrical heating	70%
Electrical heating	19%
District heating	11%
Variable contract	82%
Default contract	18%
Fixed contract	$< 1\%$

Household Clustering Using K-Means

Since there are many households in the electricity consumption data, it is relevant to not only group households by their properties, but also by their consumption patterns. The idea is to identify groups where the similarity within the group is higher than that among the groups [29]. K-Means clustering is one of the most popular algorithms for solving this problem in a simple way. K-Means is an unsupervised learning algorithm, where the main idea is to select k centroids, one for each cluster, and place them as far from each other as possible. The next step is to associate each point in the dataset with the closest centroid. After associating all points, the k centroids are recalculated by averaging the points within each cluster. This process is repeated until the centroids no longer move. Formally, the algorithm aims at minimizing the squared error function $W(S,C)$ defined in Equation 4.1 [35].

$$W(S,C) = \sum_{k=1}^{K} \sum_{i \in S_k} \|y_i - c_k\|^2 \tag{4.1}$$

In the equation, S is a partition of k clusters represented by observations in the form of vectors y_i. Each cluster S_k is non-empty and non-overlapping, and has a centroid c_k ($k = 1,2,\ldots,K$).

In these experiments, K-Means clustering is applied to define customer groups based on their electricity consumption. The elbow method is used to find an optimal number of k, i.e. the number of clusters. The elbow method is a visual method, where different values of k are plotted against the variance within the clusters [36]. As the value of k increases, the average variance becomes smaller. The goal is to find a value of k for which the variance stops decreasing sharply, that is, the elbow of the graph. Figure 5 shows the elbow plot for determining the optimal number of customer groups, in this case $k = 3$. This results in three customer groups; low consumption, medium consumption, and high consumption. Each customer in the dataset is assigned a label based on the cluster it belongs to, and this label becomes an input feature for the forecasting models. Table 3 details each cluster regarding size and consumption patterns. The large standard deviation present in each cluster suggests that outlier months with high consumption exist for all identified consumption patterns.

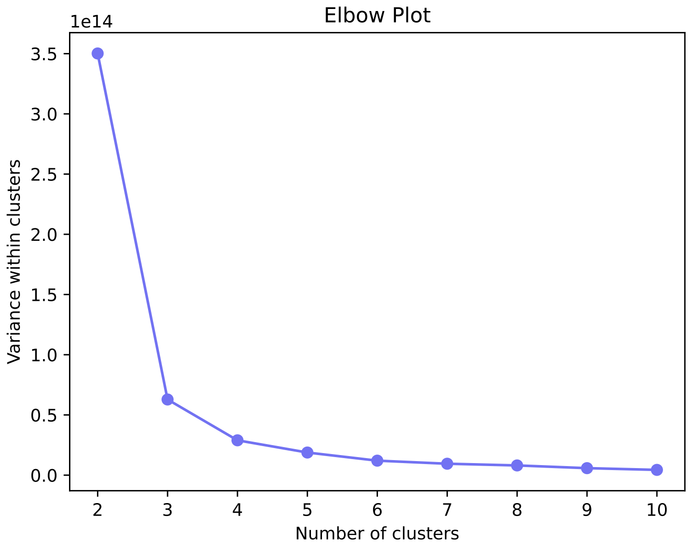


Figure 5: Elbow plot for finding the optimal number of customer profile clusters k.

Table 3 Clustered consumption profiles in the Umeå Energi dataset.

Cluster	Households	Mean (kWh)	Median (kWh)	SD (kWh)
Low consumption	1 804	365	173	515
Medium consumption	4 458	410	227	549
High consumption	7 305	524	230	724

Variable Interactions

Interactions between the variables are investigated using a correlation map, as shown in Figure 6. This correlation map includes the internal historical consumption features and SI, as well as external features in the form of weather indices (THI, WCT [see section x]) and calendar information (number of holidays, number of days). The three historical consumption variables show a high correlation with each other, while they have a low correlation with all the external variables. [conclusions from this?]

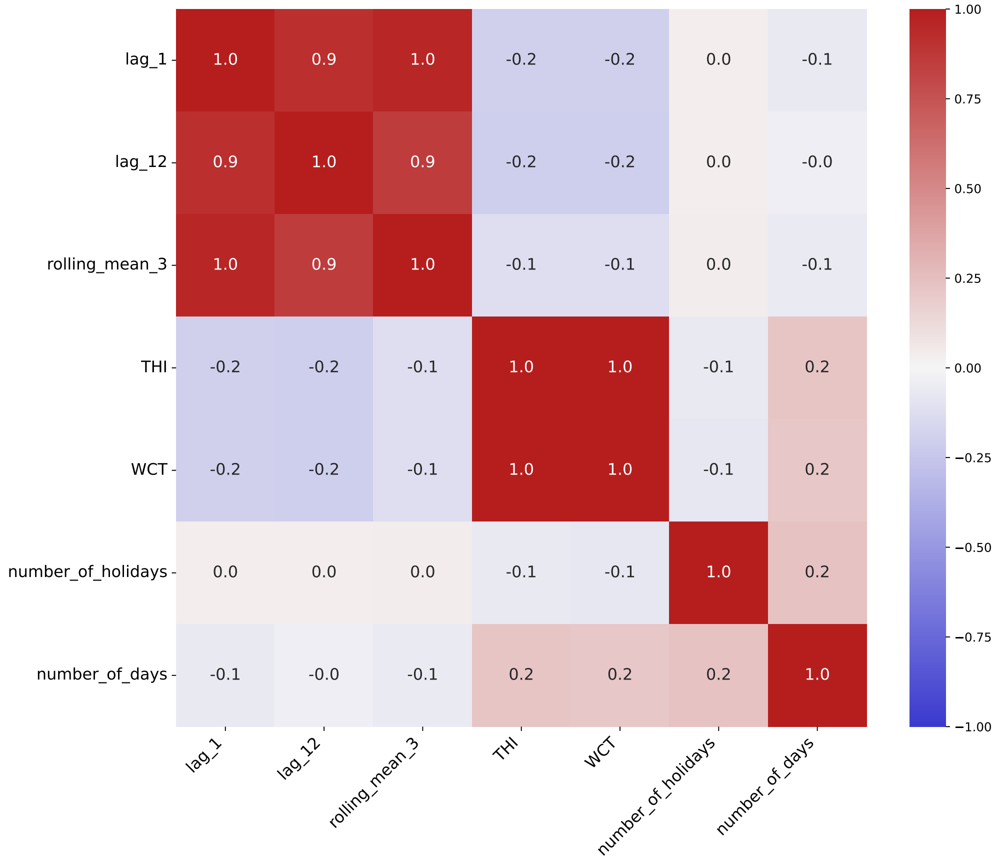


Figure 6: Correlation map of variables [placeholder].

4.1.2 External Variables

Along with the historical electricity consumption data from Umeå Energi, weather variables are retrieved from SMHI[1] and calendar information is retrieved from Dagsmart API[2]. These external variables are intended to improve the forecasts by including environmental factors that may affect consumption patterns.

Weather Data

Weather data is retrieved from SMHI's open data for the January 1st 2022 to November 30th 2025 time period. Nine variables are selected from the Umeå Airport measurement station, while three variables pertaining to solar attributes are retrieved from the Umeå Sol station, and snow depth is retrieved from Umeå Röbäcksdalen. All variables are obtained

[1]Swedish Meteorological and Hydrological Institute, SMHI. https://www.smhi.se/data (Accessed 2026-03-17)

[2]Swedish calendar data API, Dagsmart API. https://dagsmart.se/api/ (Accessed 2026-03-16)

as hourly measurements, except snow depth which is retrieved at daily intervals. Table 4 details the weather variables and their measurement units. Across all variables, 4.9% of the measurements are missing, where most missing values occur in the lowest cloud base variable that has 42% missing values.

The measurements of all variables are aggregated to monthly values using mean. Figure 7 shows the monthly values of each weather variable plotted over the entire time range. [maybe put plot in appendix?] [need to discuss how these values are aggregated (avg vs sum or other way)]. The seasonal component is apparent for many of the weather variables, while some variables such as wind speed and air pressure lack a seasonal pattern. [conclusions from this]

Table 4 Weather variables retrieved from SMHI.

Variable	**Description**	**Resolution**
temperature	Air temperature (°C)	Hourly
dew_point_temp	Dew point temperature (°C)	Hourly
snow_depth	Snow depth (m)	Daily
humidity	Relative humidity (%)	Hourly
wind_direction	Wind direction (°C)	Hourly
wind_speed	Wind speed (m/s)	Hourly
lowest_cloud	Lowest cloud base (m)	Hourly
global_radiation	Global irradiance (Sweden) (W/m^2)	Hourly
longwave_radiation	Long wave irradiance (W/m^2)	Hourly
sunshine	Sunshine amount (s)	Hourly
air_pressure	Air pressure (hPa)	Hourly
visibility	Visibility (m)	Hourly

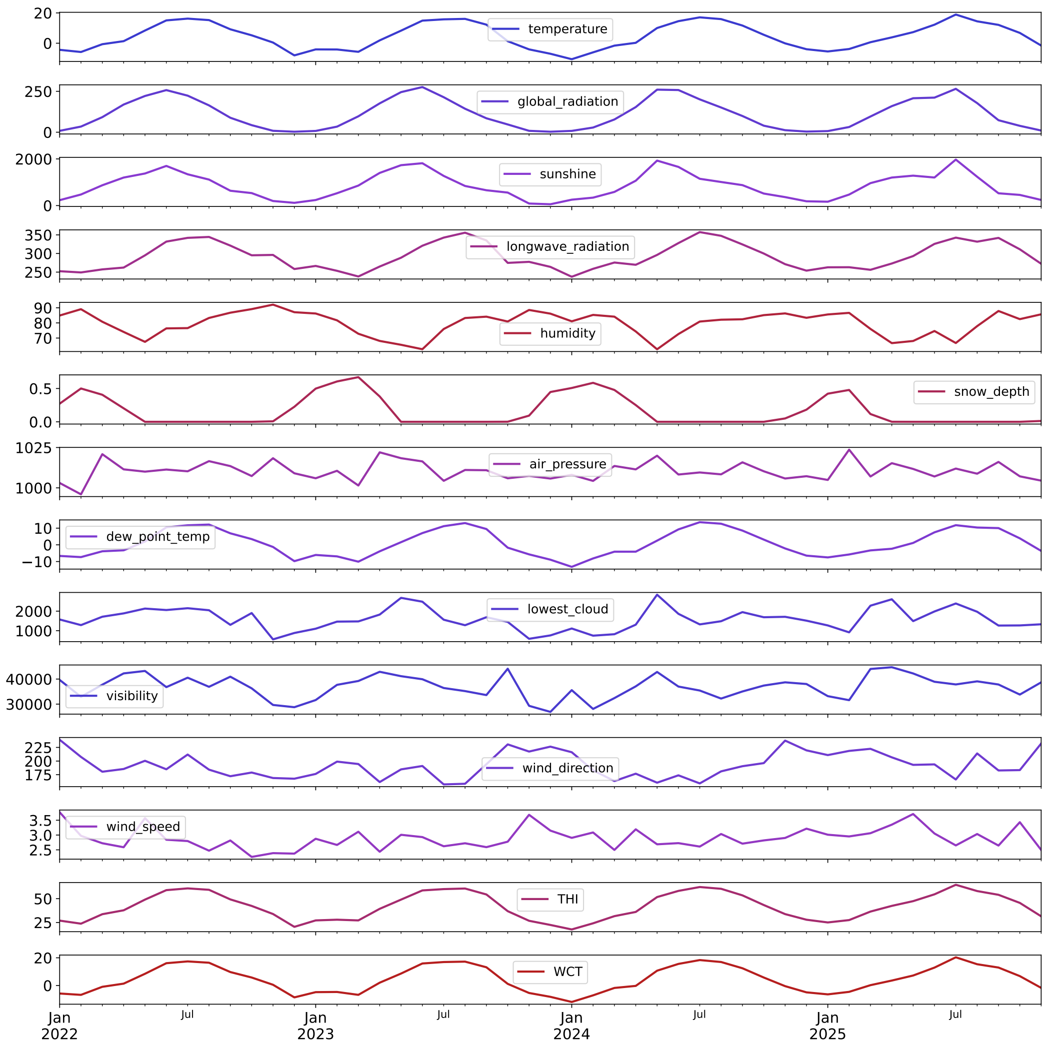


Figure 7: Weather variables' monthly values between January 2022 and November 2025.

Could be relevant to include some variable(s) of minimum and maximum temperature? As in [10]

Although weather factors have been proven to be useful for predicting electricity consumption, the link between weather and electricity consumption can be indistinct. Weather indices incorporating temperature, humidity, and wind speed can provide a better representation of the perceived environmental effects and the impact on electricity consumption [9]. Two indices, temperature humidity index (THI) and wind chill temperature (WCT), are calculated as in Equation 4.2 and Equation 4.3. In the equations, *temp* is temperature, *humi* is humidity, and *ws* is wind speed. These indices are included as input features for the forecasting models.

$$THI = (1.8 \times temp + 32) - [(0.55 - 0.0055 \times humi) \times (1.8 \times temp - 26)] \tag{4.2}$$

$$WCT = 13.12 + 0.6215 \times temp - 11.37 \times ws^{0.16} + 0.3965 \times temp \times ws^{0.16} \tag{4.3}$$

Calendar Information

In addition to weather data, calendar information is included as external factors. During holidays, energy consumption behaviors change due to differences in social patterns. Energy consumption is not only dependent on the decision-making of individuals, it also relies on the cultural context of social and economic activities [37]. Holiday information is retrieved from Dagsmart, an open Swedish calendar information API. All holidays and weekend days during the years 2022 through 2025 are obtained, and these are compiled to a *number of holidays* feature that indicates the number of holidays for each month in the dataset. In Figure 8 the number of holidays per month is visualized, showing that December is month with the most holidays. The fewest holidays are during the rest of the winter and autumn months. An additional *number of days* feature is constructed for each month in the dataset, since the number of days varies (e.g., February is missing several days of consumption compared to months with 31 days).

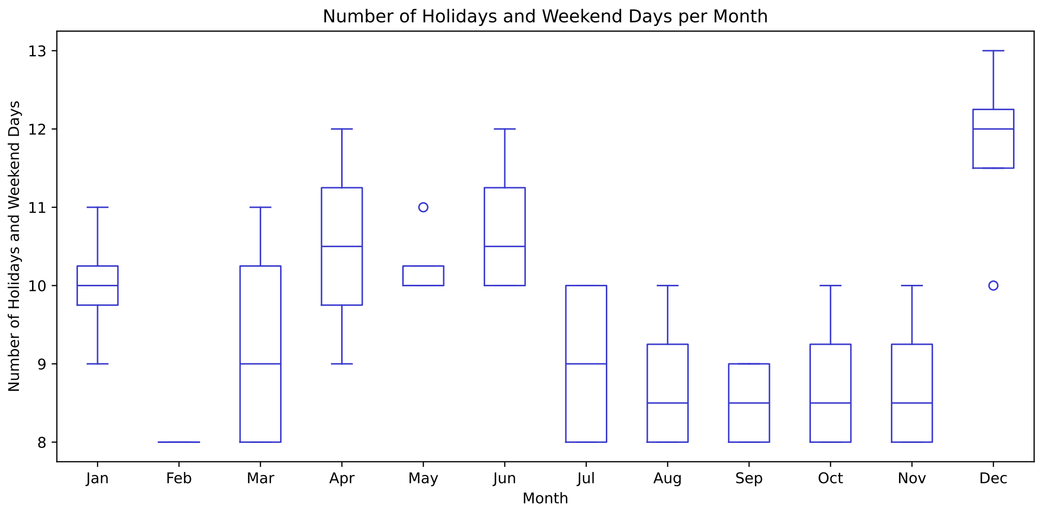


Figure 8: Number of Swedish holidays per month during the years 2022 through 2025.

4.1.3 Feature Selection

The historical consumption, weather, and calendar variables are compiled to a complete feature set for the forecasting models. A feature selection is made from the weather variables, based on the correlation maps found in Figure 19 and Figure 6. Variables with a high correlation to several other variables are excluded from the feature set, in this case global radiation, long wave radiation, dew point temperature, and visibility. This is due to highly correlated features accounting for each other's impact to the model output [formulera bättre], and the SHAP method assuming that features are uncorrelated [23]. The complete selected feature set is found in Table 10. A subset of 2 000 random households are selected for the model training and evaluation. The data is then split into training and test data such that the years 2023 and 2024 are the training set, and January through November 2025 is the test data (see Table 5).

Categorical features in the dataset are label encoded to integer values for all models except CatBoost. This means that each categorical value is represented by a unique integer, instead of its original string value. Label encoding is one of the simplest encoding techniques, but works well for tree-based models [borde göra one-hot encoding istället] [38] okej fast AI säger att label encoding är bättre för tree models, kan testa båda och skriva om det

Table 5 Train-test split of the dataset.

Data	Time range	Percentage	Data points
Training data	January 2023 to December 2024	69%	48 000
Test data	January 2025 to November 2025	31%	22 000

4.2 Ensemble Learning Forecasting Models

Forecasting is performed using different decision tree-based ensemble learning models. Decision trees classify data items by asking a sequence of questions about the items' features. The nodes of the tree contain questions, and for each possible answer, every internal node points to one child node. The sorting of an item into a class begins from the root of the tree, and according to the item's answer to the questions, it follows the path to a leaf node [39]. An example of a decision tree structure is found in Figure 9. One large advantage with decision trees is that they are considerably easier to interpret than neural networks, as they follow logical rules instead of having connections with numeric weights between nodes. Decision trees perform well for a few highly relevant attributes, but not as well if there are many attributes with complex interactions. Ensembles of decision trees can be employed to increase accuracy when the training data is more complex and changing [40]. Decision tree ensembles are considered among the most reliable and precise methods for regression and classification tasks, and they have been shown to be effective for predicting energy consumption [9]. However, one disadvantage is that the interpretability decreases when multiple classifiers are involved, making it more difficult to understand the reasoning behind the ensemble's decision [40]. This creates a need for model explainability outside the ensemble's inner workings, which in this thesis is addressed by applying explainable AI analysis.

Five decision tree-based ensemble models are selected as forecasting models, as in Moon et al.'s work [9]; random forest (RF), gradient boosting machine (GBM), extreme gradient boosting machine (XGBoost), light gradient boosting machine (LightGBM), and categorical boosting (CatBoost). [these models are selected because...]

Random forest is a decision tree ensemble learning algorithm where the trees depend on a set of random feature splits at each node. This means that every tree produces its own prediction according to its set of data. The individual predictions are then combined into a final prediction via majority voting. RF is an extension of the bagging method, efficient for large datasets and robust to outliers and overfitting, making it suitable for this application. Additionally, RF is faster than other bagging or boosting techniques [41].

Gradient boosting machines are a family of ML algorithms where the learning process consecutively fits a new tree with the goal of reducing the errors of the previous one. This is done by constructing new base-learners that are maximally correlated to a decreasing loss function. The loss function can be of various kinds, which makes GBM highly flexible for different data-driven tasks. It is up to the model designer to choose the most appropriate loss function [42]. GBM is able to learn complex patterns and interactions in the data due

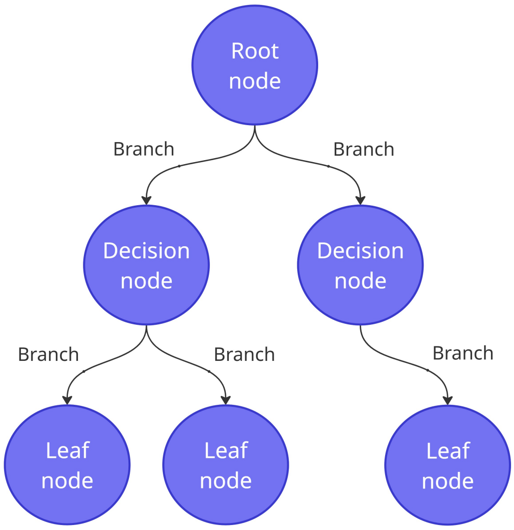


Figure 9: A decision tree.

to its iterative learning process [9].

XGBoost is a specialization of the gradient boosting algorithm with several optimizations, creating a scalable tree boosting system. The optimizations include a sparsity-aware algorithm for handling missing values, approximate learning via a weighted quantile sketch algorithm, and efficient cache access patterns reducing runtime. This results in a fast and robust boosting model using minimal resources [43].

LightGBM is another gradient boosting algorithm that focuses on efficiency for cases with many input features and when the amount of data is large. LightGBM uses two novel techniques to improve performance and efficiency, *gradient-based one-side sampling* (GOSS) and *exclusive feature bundling* (EFB). GOSS excludes data points with smaller gradients that contain less information, with the goal filtering out noisy data while retaining information gain. EFB bundles mutually exclusive features (that rarely have nonzero values at the same time) to create a smaller feature space with nearly exclusive features [44].

CatBoost is a gradient boosting algorithm that uses a technique called *ordered boosting* to reduce prediction shift caused by target leakage which occurs in other gradient boosting algorithms. CatBoost also encodes categorical features uniquely, by applying a technique called *ordered target statistics* to handle high cardinality features [45]. Because of this, CatBoost is able to achieve high accuracy, especially when the data contains a significant number of categorical features [9].

[maybe specify advantages and disadvantages with these models more clearly, like moon et al. might also have to justify why I use these models more]

4.2.1 Configuration of Models

To configure the models, a search for the best hyperparameters is performed via exhaustive grid search. The optimization consists of a five-fold time series cross-validation process over the hyperparameters specified in Table 6, using the scikit-learn packages GridSearchCV and TimeSeriesSplit. A subset of 30% of the training dataset is used for the grid search. For XGBoost, the grid search is minimized to 50 random combinations of the search space, in order to reduce execution time.

Table 6 Hyperparameters for decision tree ensemble models.

Model	**Hyperparameters**
RF	Trees count: 128 Features per split: auto, sqrt, log2
GBM	Iteration count: 100, 250, 500 Learning rate: 0.01, 0.05, 0.1 Depth: 5, 10 Loss type: quantile, Huber
XGBoost	Iterations: 250, 500, 1000 Learning rate: 0.01, 0.05, 0.1 Depth: 6, 8, 10 Subsampling rate: 0.5, 0.75, 1.0 Feature sample by tree/level/node: 0.5, 0.75, 1.0 Booster type: gbdt, dart
LightGBM	Iterations: 1000, 1500 Learning rate: 0.01, 0.05, 0.1 Leaves: 64 Subsample: 0.5 Feature sample by tree: 1.0 Booster type: gbdt, dart
CatBoost	Learning rate: 0.03, 0.1 Max tree depth: 4, 6, 10 L2 regularization levels: 1, 3, 5, 7, 9

4.3 Evaluation Metrics

To evaluate the predictive performance of the forecasting models, three statistical metrics are used; mean absolute percentage error (MAPE), root mean squared error (RMSE), and mean absolute error (MAE). MAPE is one of the most common metrics to evaluate forecasting accuracy, due to its scale-independence and interpretability. It defines the average of the absolute percentage errors, as in Equation 4.4. One disadvantage to be aware of is that MAPE yields infinite or undefined values when the actual values are zero or close to zero [46]. This experiment handles this by removing any households from the dataset that have less than 10 kWh consumption during some month. RMSE and MAE, as defined in Equation 4.5 and Equation 4.6, are utilized to give insights on the size of the errors in kWh. These two metrics are also among the most commonly used in the electricity consumption forecasting field [47].

$$MAPE = \frac{1}{n} \times \sum \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \tag{4.4}$$

$$RMSE = \sqrt{\frac{1}{n} \times \sum (y_t - \hat{y}_t)^2} \tag{4.5}$$

$$MAE = \frac{1}{n} \times \sum |y_t - \hat{y}_t| \tag{4.6}$$

In the equations, y_t denotes the actual observed value at time t, while $\hat{y}_t$ denotes the predicted value at time t. The total number of prediction instances is denoted by n.

4.4 Explainable AI Analysis

To enhance the interpretability of the forecasts and investigate the influence of model features, a SHAP analysis is performed. The SHAP methodology [Explain SHAP for ensemble models specifically, like in [9], explain why SHAP gives better insights than just feature importances (https://machinelearningmastery.com/a-gentle-introduction-to-shap-for-tree-based-models/)] [beeswarm plot, feature importance plot]

This experiment uses the SHAP *TreeExplainer*, specifically designed for decision tree models.

4.5 Experimental Setup

Experiments are performed in the Python language. To process and analyze the data, the *pandas*, *seaborn*, and *matplotlib* packages are utilized. The *scikit-learn* package provides the RF and GBM models, as well as GridSearchCV for the exhaustive grid search. XGBoost, LightGBM, and CatBoost models are obtained from [deras egna repon typ?]. The *shap* Python package is used for the explainable AI analysis.

Experiments are run in a Windows virtual environment with a [GPU]. GPU acceleration is used when performing grid search, training models, and performing SHAP analysis.

5 Results

[some introductory text to the chapter]

5.1 Performance of Forecasting Models

The performance of the five decision tree-based ensemble models is assessed via MAPE, RMSE, and MAE metrics. Table 7 shows the scores for each model, separated by internal features and total features. Scores under the internal feature category represent the models' performance when only trained using the historical electricity consumption, while total features refers to the combination of internal and external features. The best performing model is XGBoost, with 14.96% errors on the total feature set. RF shows the worst performance with 16.33% errors and considerably higher RMSE and MAE scores on the total feature set, compared to the other models.

Results indicate that the inclusion of external factors does help with improving the forecast accuracy in terms of MAPE, but that MAE and especially RMSE scores may actually worsen, a pattern that is apparent among all models. MAPE is more sensitive to errors on small consumption values, while RMSE is more sensitive to high consumption values and outliers. This suggests that the external factors help with increasing forecast accuracy for households with a lower and more stable consumption, while households with a high or varied consumption do not benefit from the external factors.

All models show a large discrepancy between RMSE and MAE scores in terms of kWh, suggesting that the dataset contains many large outliers [move to discussion possibly]. Since RMSE squares the errors before averaging, large errors are penalized more, compared to MAE where all errors are treated equally via the absolute value.

Table 7 Performance of forecasting models.

	Internal features			**Total features**		
Model	**MAPE (%)**	**RMSE (kWh)**	**MAE (kWh)**	**MAPE (%)**	**RMSE (kWh)**	**MAE (kWh)**
RF	16.34	142.47	65.93	16.33	171.47	74.68
GBM	15.67	144.07	64.49	15.07	154.97	66.67
XGBoost	15.40	144.12	63.94	14.96	153.55	65.46
LightGBM	15.67	150.21	66.09	15.18	159.54	67.03
CatBoost	15.73	152.85	65.74	16.05	150.07	66.52

A pattern occurring for all models is that on average, the models overestimate the electricity consumption during the winter months in the beginning of the year, as illustrated in Figure 10. During the summer, the average predictions lie closer to the actual average consumption of the households. The scatter plots in Figure 11 show that the overestimation of electricity consumption is a general trend for all models except XGBoost, indicated by the regression lines with larger slopes than the perfect prediction line. For those models, it is clear that predictions in the several thousand kWh range are commonly overestimated, while the errors for lower kWh predictions are more evenly distributed around the perfect prediction line.

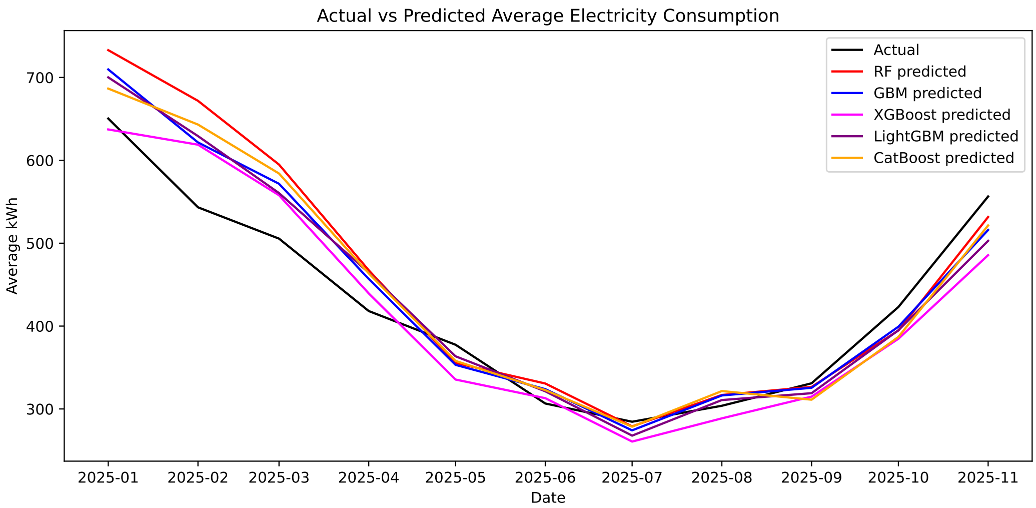


Figure 10: Average actual vs predicted electricity consumption for all models.

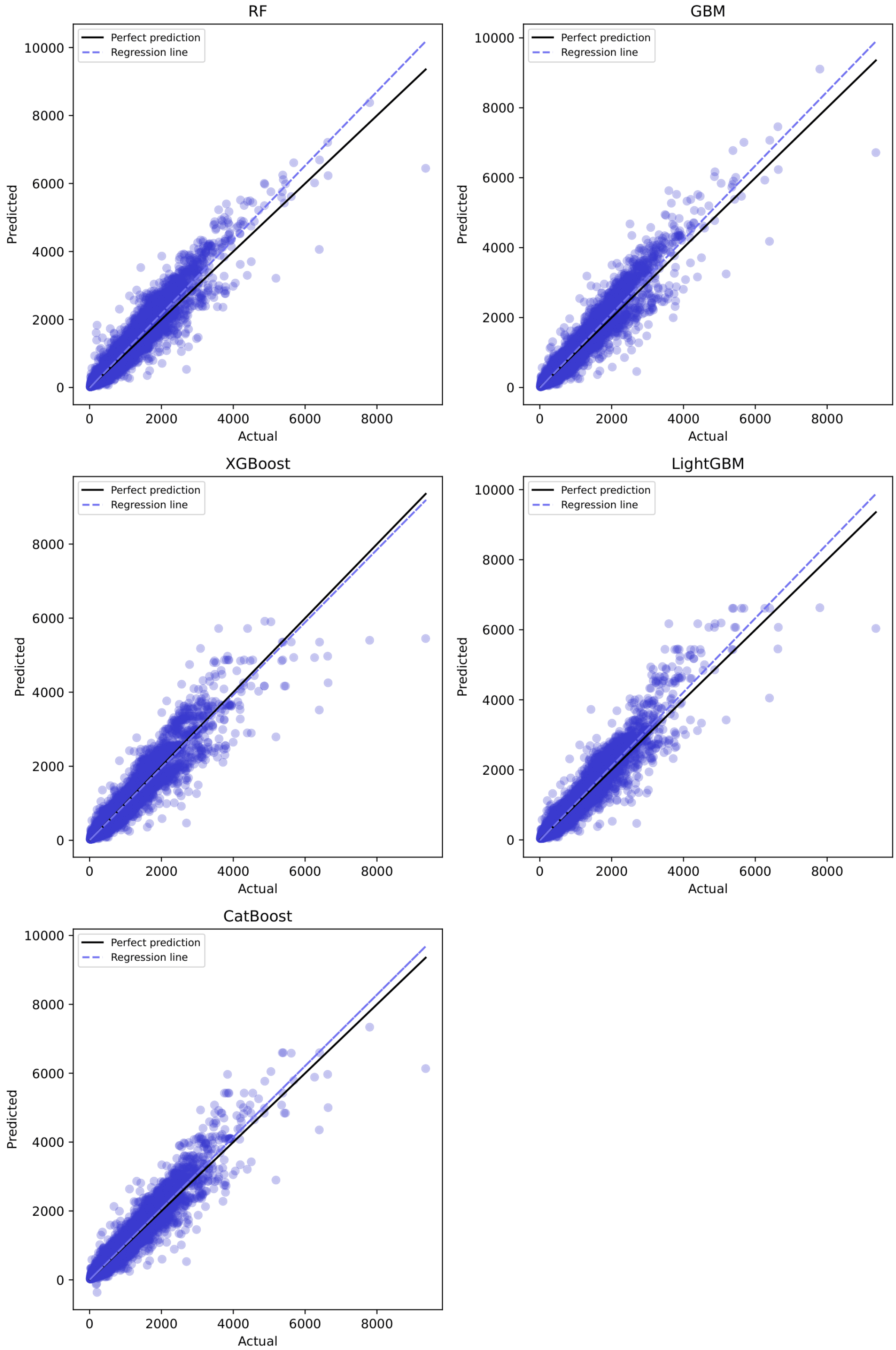


Figure 11: Scatter plots showing prediction error distribution for each model.

5.1.1 Comparison with Baseline Model

The baseline linear regression model used for comparison uses historical consumption and temperature as input variables. The past 24 months of electricity consumption and monthly average temperature is used to predict the following month's consumption. Evaluation of the linear regression model was performed using the same 2000 household sample as for the ensemble models, and the results are found in Table 8. The linear regression model performs worse than the best ensemble models by roughly 3.5% in percentage errors. RMSE and MAE scores are also worse for the linear model, especially when compared to ensemble models using only internal features.

Table 8 [].

Model	MAPE (%)	RMSE (kWh)	MAE (kWh)
Linear regression	18.47	161.43	71.98

[maybe try LSTM and ANN for comparison too]

5.2 Explainable AI Analysis of Forecasts

All five ensemble learning models are analyzed via the SHAP method. Firstly, the models were trained using their optimal hyperparameters, and feature importance metrics were retrieved for the trained models. Secondly, a tree SHAP analysis was performed to calculate the Shapley values for all features. For the RF model, approximate SHAP values were used due to the full calculation being time-consuming for this particular model. Figure 12 - Figure 16 show the resulting feature importance plots (left) and SHAP summary plots (right) [change to a) and b)] for each model.

Both the feature importance and SHAP analysis show that the lag_1 and lag_12 historical consumption features are highly influential for all models. The SHAP summary plots, showing the effect of individual feature values on the prediction, demonstrate that these features have a clear relationship where low feature values contribute to a lower predicted value, and high feature values contribute to higher predictions. Most models prioritize the lag_1 feature over lag_12 in their decision-making, except for RF. The $rolling\_mean\_3$ feature is ranked relatively high for all models, but the SHAP summary plots show that the variable may have a more complex relationship to the model output in some models. For GBM, XGBoost, and LightGBM, high values of the $rolling\_mean\_3$ feature contribute both to decreasing and increasing predicted values.

Interestingly, for the XGBoost model, the WCT feature is ranked the highest in feature importance. However, the more detailed SHAP summary plot shows that lag_1 and lag_12 are the most influential, while WCT is ranked in the middle.

Among the external variables, wind direction and sunshine amount are consistently ranked highly for all models.

The two calendar information features, $number\_of\_holidays$ and $number\_of\_days$, are commonly ranked among the less important features for the models, along with the *year* timestamp variable. The *month* variable, indicating the month of the year, is however ranked highly for several models.

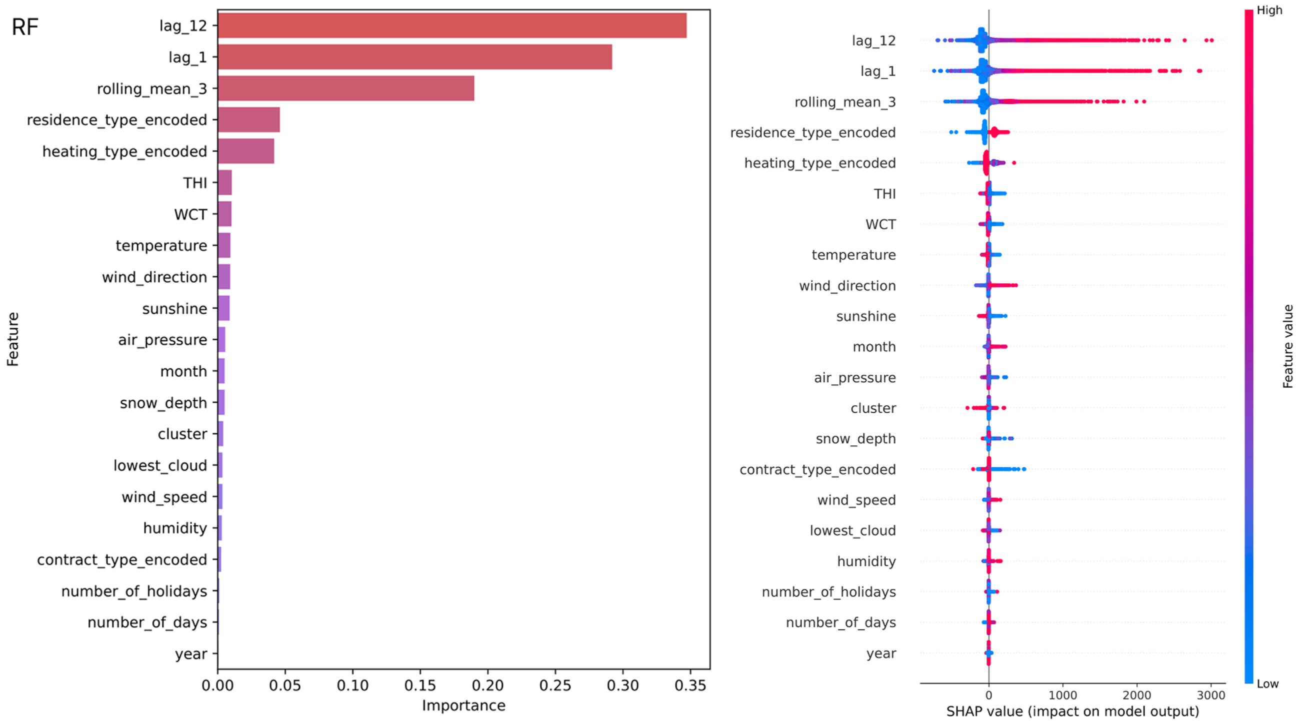


Figure 12: Feature importance and SHAP summary for RF.

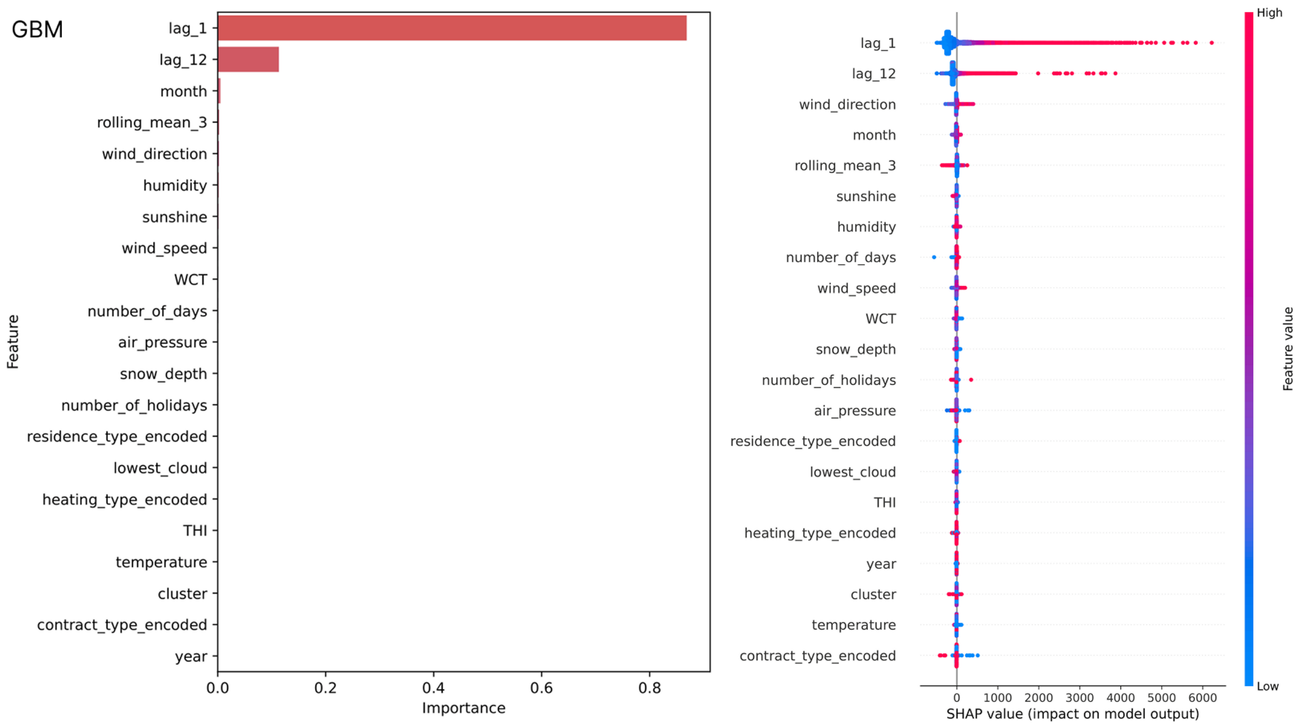


Figure 13: Feature importance and SHAP summary for GBM.

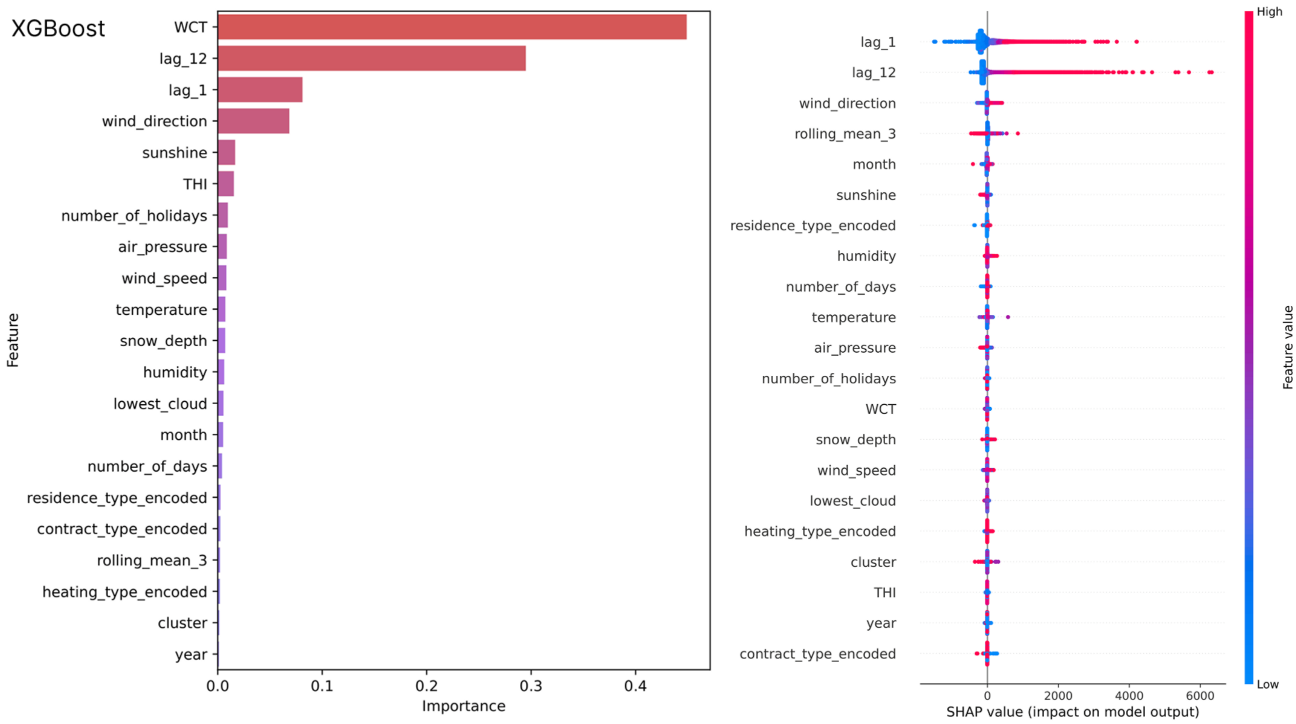


Figure 14: Feature importance and SHAP summary for XGBoost.

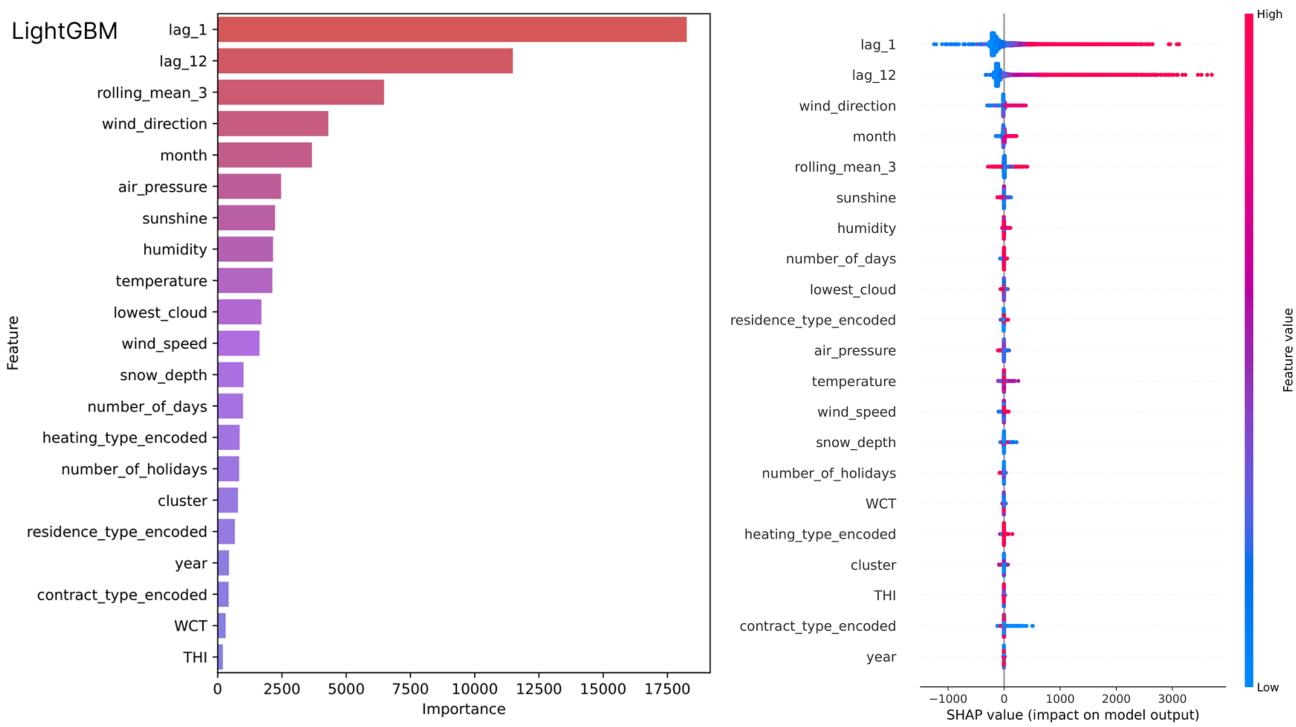


Figure 15: Feature importance and SHAP summary for LightGBM.

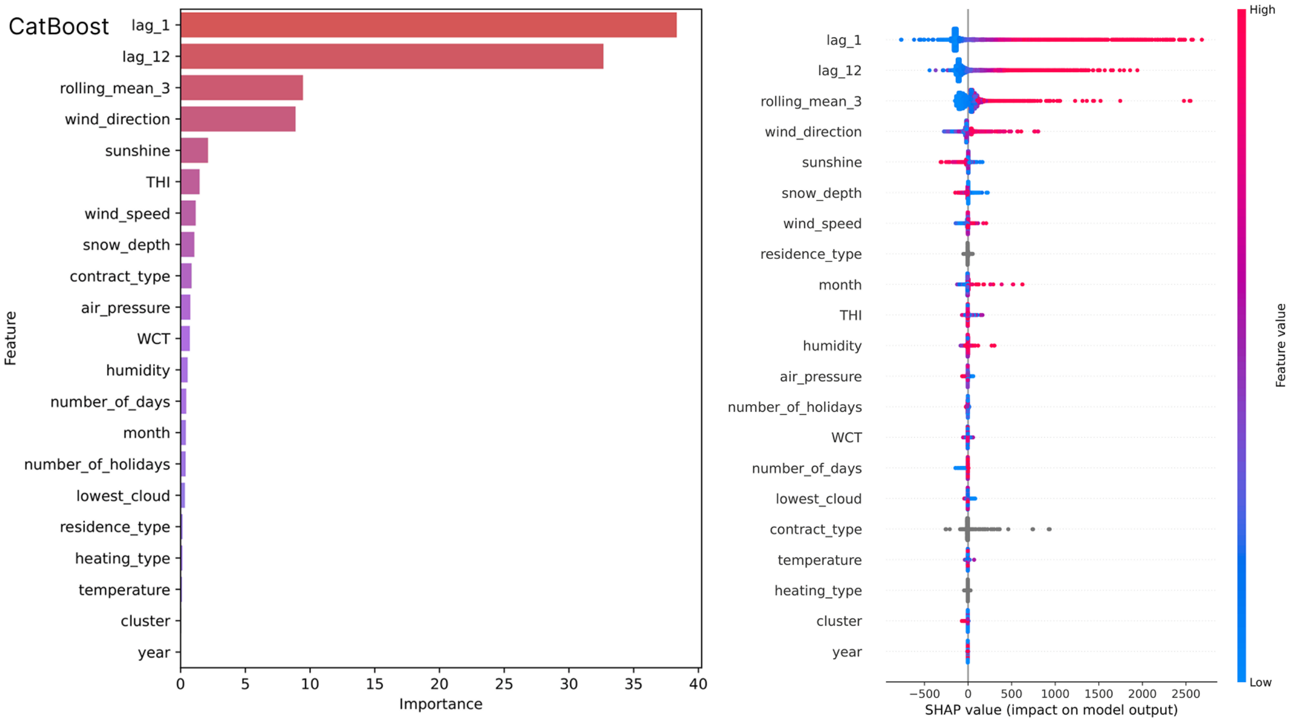


Figure 16: Feature importance and SHAP summary for CatBoost.

6 Discussion

[holiday indicators are probably more useful for short-term forecasting (like daily forecasting, you would know if you are forecasting a holiday or a weekday which may be important. but this effect smooths out over an entire month)]

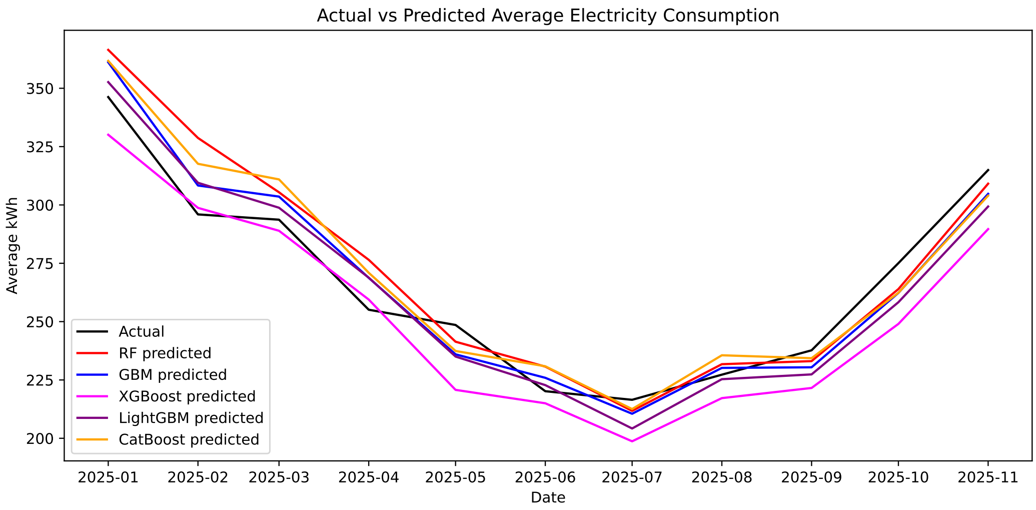
[household energy prediction is unique because of outliers, the same problem is not prominent when predicting for an entire city e.g. future research is (probably) needed to better handle the outlier situation of household prediction, since this topic is underexplored in research]

To investigate the effect of outliers, further analysis and experiments were performed. A maximum monthly consumption threshold of 2000 kWh was selected, due to the majority of observations registering below 2000 kWh, as shown in Figure 2. 267 households exceeding this threshold were removed from the sample of 2000 households used for the experiments. The grid search, training, and evaluation process was repeated for each model with the modified data. The resulting scores are found in Table 9.

The removal of high-consuming households results in a small improvement in MAPE, and large improvements in both RMSE and MAE, confirming that high-consuming households have a great impact on the models' RMSE and MAE performance. The root mean squared errors were roughly half of the original for all models, and mean absolute errors decreased with around y kWh. MAPE scores improved up to 1 percentage point. Notably, the RMSE and MAE scores for the total feature group (both internal and external features) are better than for the internal group when the high-consuming households have been removed. [This suggests that external variables create noise for the households with high consumption, and actually worsen the predictive performance for this group. However, it is clear that the external variables improve the accuracy for households with a lower or medium consumption (vill förtydliga att det troligen bara är de med extremt hög konsumtion som förstör?)]

Table 9 Performance of forecasting models when outlier households (i.e., households with >2000 kWh consumption any month) are removed from the data.

Model	**Internal features**			**Total features**		
	MAPE (%)	**RMSE (kWh)**	**MAE (kWh)**	**MAPE (%)**	**RMSE (kWh)**	**MAE (kWh)**
RF	15.99	70.67	38.37	15.32	69.42	37.03
GBM	15.27	70.41	37.29	14.43	66.85	35.11
XGBoost	15.25	70.17	37.74	14.40	68.52	36.27
LightGBM	15.38	72.44	38.52	14.37	66.47	35.30
CatBoost	15.44	70.03	37.34	15.12	66.05	35.53



Actual vs Predicted Average Electricity Consumption
Average kWh
350
325
300
275
250
225
200
Actual
RF predicted
GBM predicted
XGBoost predicted
LightGBM predicted
CatBoost predicted
2025-01
2025-02
2025-03
2025-04
2025-05
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
Date

Figure 17: .

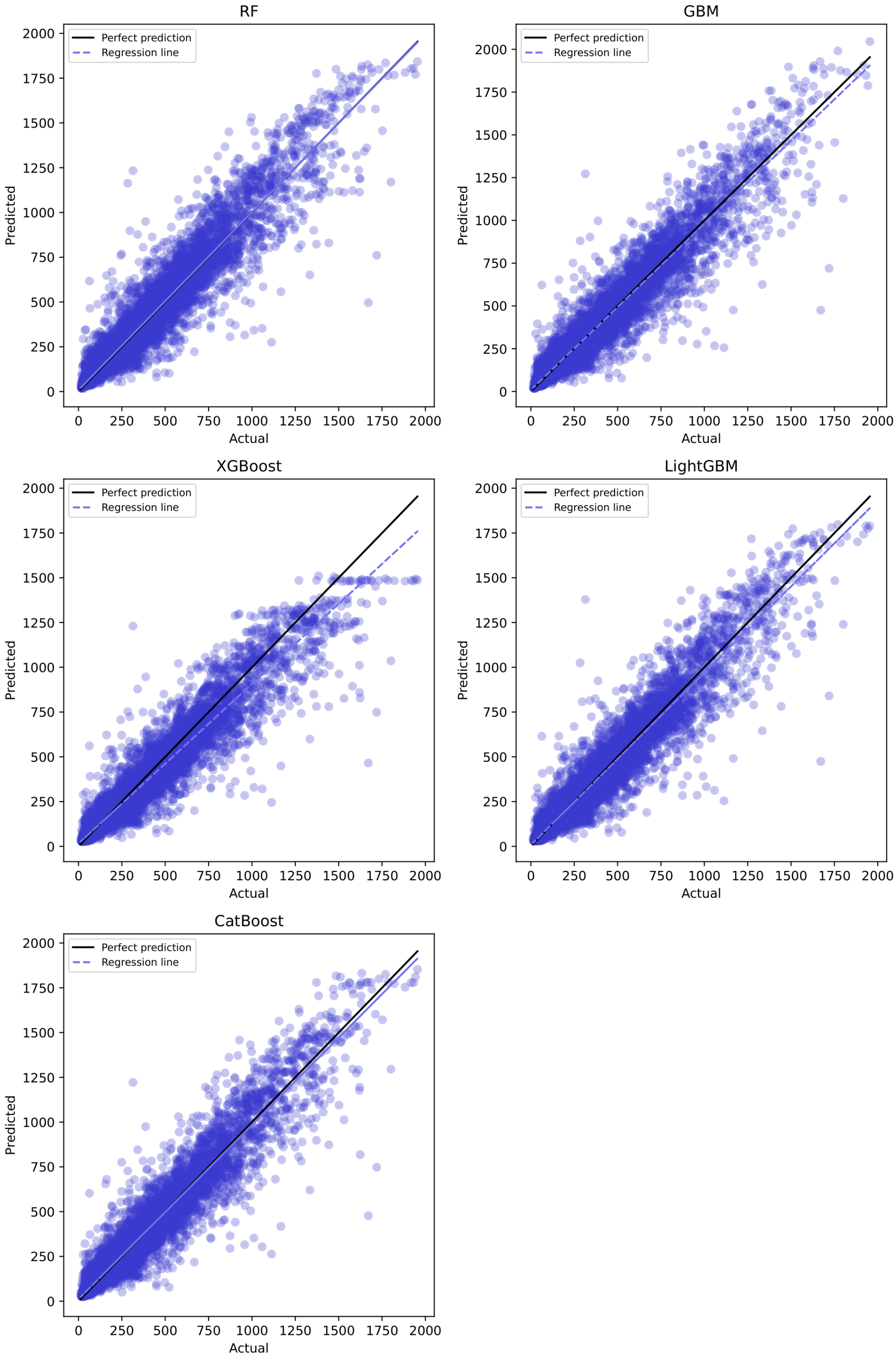


Figure 18: .

TODO: analysera mer om dessa outliers. alla bör ju ha eluppvärmning tex?

The removed households largely belong to the house category (89%) and vacation home category (10%), and the majority have electrical heating (70%).

[discuss how SHAP gives better insights than FI, XGBoost example]

[limitations]

7 Conclusion

7.1 Future Work

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A First Appendix

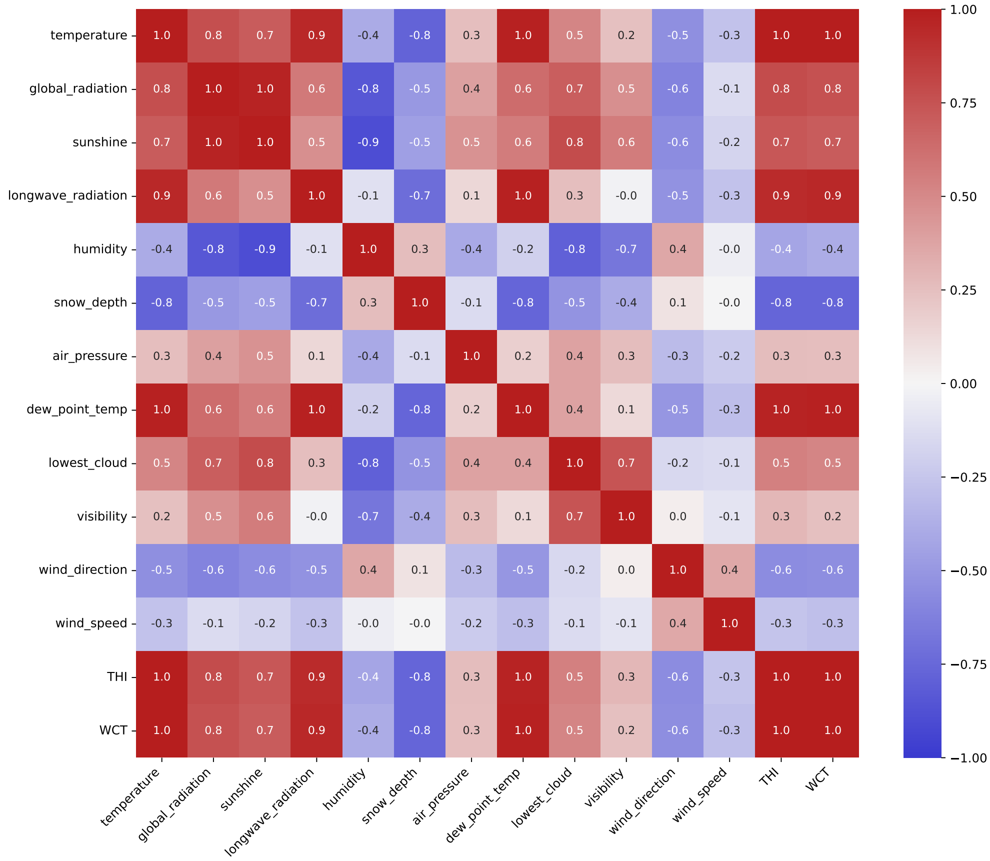

Figure 19: Correlation map of weather variables [placeholder].

Table 10 Selected input variables.

Variable	Description	Data Type
month	Month of the year	Numeric
year	Year	Numeric
number_of_holidays	Number of holidays and weekend days during month	Numeric
number_of_days	Number of days during month	Numeric
cluster	Customer cluster	Customer information (categorical)
building_type	Type of building	Customer information (categorical)
heating_type	Type of heating system	Customer information (categorical)
contract_type	Type of contract	Customer information (categorical)
temperature	Monthly average air temperature (°C)	Weather condition (numeric)
snow_depth	Monthly average snow depth (m)	Weather condition (numeric)
humidity	Monthly average relative humidity (%)	Weather condition (numeric)
wind_direction	Monthly average wind direction (°C)	Weather condition (numeric)
wind_speed	Monthly average wind speed (m/s)	Weather condition (numeric)
lowest_cloud	Monthly average lowest cloud base (m)	Weather condition (numeric)
sunshine	Monthly average sunshine amount (s)	Weather condition (numeric)
air_pressure	Monthly average air pressure (hPa)	Weather condition (numeric)
THI	Temperature humidity index	Weather index (numeric)
WCT	Wind chill temperature index	Weather index (numeric)
*lag_*1	Consumption for the previous month (kWh)	Historical consumption (numeric)
*lag_*12	Consumption for the same month last year (kWh)	Historical consumption (numeric)
*rolling_mean_*3	Average consumption for the previous three months (kWh)	Historical consumption (numeric)